

HealthCare Project- SQL

Total Number of Patients

```
SELECT  
    COUNT(*) AS total_patients  
FROM  
    healthcare;
```

	total_patients
▶	54966

Count patients by gender

```
SELECT
    gender, COUNT(*) AS total
FROM
    healthcare
GROUP BY gender;
```

	gender	total
▶	Male	27496
	Female	27470

Average Age of Patients

```
SELECT  
    ROUND(AVG(age), 0) AS avg_age  
FROM  
    healthcare;
```

	avg_age
▶	52

Most common Medication

```
SELECT
    Medication, COUNT(*) AS total
FROM
    healthcare
GROUP BY Medication
ORDER BY total DESC
LIMIT 1;
```

	Medication	total
►	Lipitor	11038

Number of Patients Per Hospital (Limited to First 10 Records.)

```
SELECT
    Hospital, COUNT(*) AS total_patients
FROM
    healthcare
GROUP BY Hospital
ORDER BY total_patients DESC ;
```

Hospital	total_patients
LLC Smith	44
Ltd Smith	39
Johnson PLC	37
Smith Ltd	37
Smith PLC	36
Smith Group	36
Johnson Inc	34
Sons Smith	33
Smith Inc	33
Smith LLC	32

Average billing amount by gender

```
SELECT
    Gender, ROUND(AVG(billing_amount), 0) AS avg_bill
FROM
    healthcare
GROUP BY Gender;
```

Gender	avg_bill
Male	25614
Female	25474

Top 5 most expensive treatments

```
SELECT
    medical_condition, ROUND(AVG(billing_amount), 0) AS avg_cost
FROM
    healthcare
GROUP BY medical_condition
ORDER BY avg_cost DESC
LIMIT 5;
```

medical_condition	avg_cost
Obesity	25804
Diabetes	25660
Asthma	25633
Arthritis	25512
Hypertension	25503

Monthly Patients Admission

```
SELECT
    MONTH(admission_date) AS month, COUNT(*) AS total_patients
FROM
    healthcare
GROUP BY month
ORDER BY month;
```

month	total_patients
1	4655
2	4210
3	4622
4	4478
5	4555
6	4650
7	4765
8	4785
9	4508
10	4613
11	4508
12	4617

Rank Hospitals By Revenue (Limited to First 10 Records.)

```
select Hospital,  
sum(billing_amount) as total_revenue,  
rank() over (order by sum(billing_amount) desc) as rank_hospital  
from healthcare  
group by hospital  
Limit 10;
```

Hospital	total_revenue	rank_hospital
Johnson PLC	1081477	1
LLC Smith	1030192	2
Smith PLC	1029426	3
Ltd Smith	1003369	4
Smith Ltd	970037	5
Johnson Inc	934310	6
Group Smith	902975	7
Inc Brown	877960	8
LLC Johnson	816439	9
Smith Group	806636	10

Insurance Provider Impact

```
SELECT
    insurance_provider, SUM(billing_amount) AS total_revenue
FROM
    healthcare
GROUP BY insurance_provider
ORDER BY total_revenue DESC;
```

insurance_provider	total_revenue
Cigna	284334110
Medicare	282911090
Blue Cross	280409098
UnitedHealthcare	279915272
Aetna	276498700

Doctor Performance Ranking

```
select Doctor,  
sum(billing_amount) as total_revenue,  
rank() over (order by sum(billing_amount) desc) as rank_doctor  
from healthcare  
group by Doctor  
Limit 10;
```

Doctor	total_revenue	rank_doctor
Michael Smith	784501	1
John Smith	610110	2
Robert Smith	609161	3
Robert Johnson	524192	4
David Smith	473346	5
William Johnson	469198	6
John Johnson	466769	7
James Smith	461820	8
Michael Johnson	460819	9
David Johnson	445604	10

Most Profitable Medical Condition

```
SELECT
    medical_condition, SUM(billing_amount) AS total_revenue
FROM
    healthcare
GROUP BY medical_condition
ORDER BY total_revenue DESC;
```

medical_condition	total_revenue
Diabetes	236486929
Obesity	236006680
Arthritis	235167560
Hypertension	233378544
Asthma	233136353
Cancer	229892204

Resource Utilization Analysis (Limited to first 10 Records.)

```
SELECT
    Name,
    Age,
    Gender,
    medical_condition,
    Hospital,
    stay_days,
    billing_amount
FROM
    healthcare
WHERE
    stay_days > (SELECT
        AVG(stay_days)
    FROM
        healthcare)
    AND billing_amount < (SELECT
        AVG(billing_amount)
    FROM
        healthcare)
    order by stay_days desc
limit 10;
```

Name	Age	Gender	medical_condition	Hospital	stay_days	billing_amount
Terry Wilson	43	Male	Diabetes	Medina Elliott Stewart	30	-1317
Lisa Todd	65	Male	Hypertension	Johnson-Bryan	30	3056
Christine Curtis	66	Male	Diabetes	LLC Woods	30	23332
Matthew Monroe	70	Female	Hypertension	Johnson-Miller	30	20823
Timothy Reyes	43	Female	Diabetes	Sons Montgomery	30	21355
William Stephens	57	Female	Hypertension	Group Harris	30	2747
Rachel Lang	44	Female	Cancer	Gray-Bond	30	21289
Melanie Johnson	32	Male	Cancer	Garcia-Norton	30	17663
Angela Nelson	85	Female	Cancer	Allen Taylor Bennett	30	14843
Brian Nichols	51	Female	Arthritis	Alexer-Moore	30	12065

The background is a light blue gradient. It features several faint, stylized icons: first aid kits with white crosses and hands holding hearts with white crosses. There are also white wavy lines scattered across the background.

Thank You